

Bachelor AI - MLOps

15 FTI PIPELINE ARCHITECTURE

Tobias Mérinat

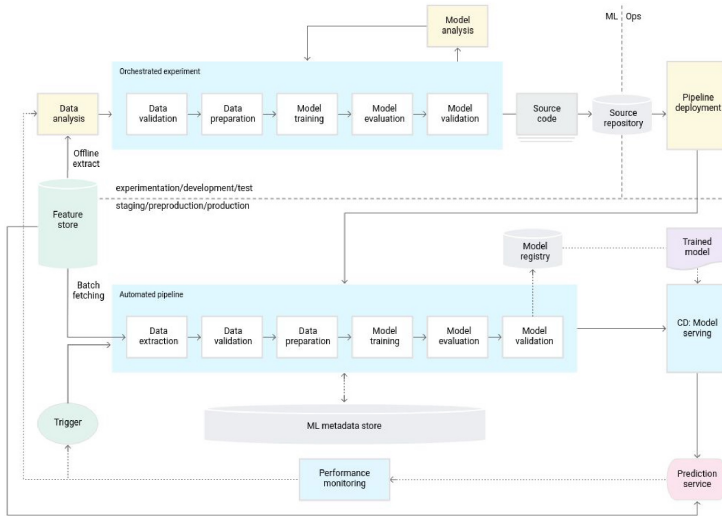
`tobias.merinat@hslu.ch`



DEPARTMENT OF INFORMATION TECHNOLOGY
Lucerne University of Applied Sciences and Arts
6343 Rotkreuz, Switzerland

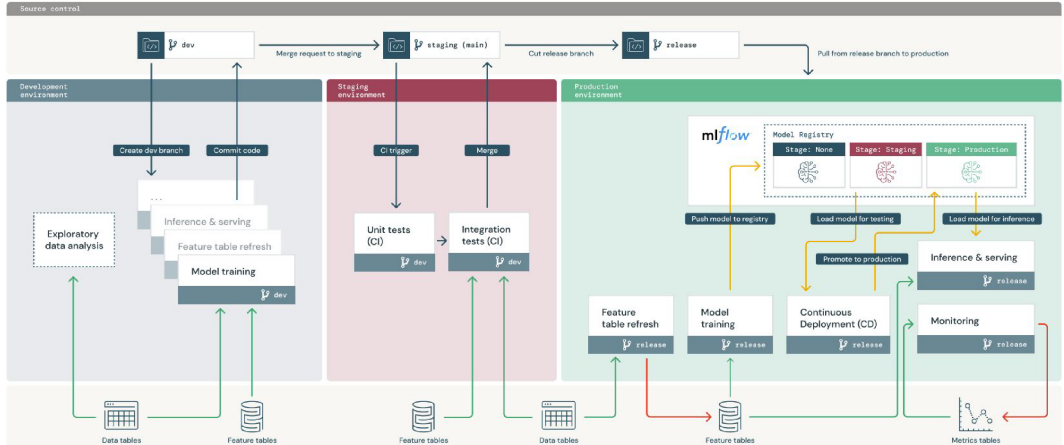
September 17/24 and October 15 2026

Architecture Overview 1/2



Source: Google

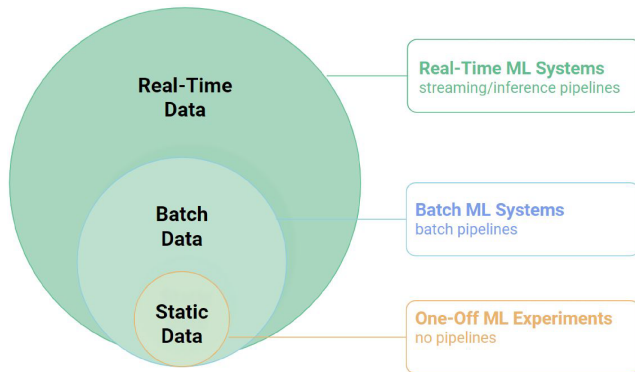
Architecture Overview 2/2



Source: Databricks

Experiment vs. System

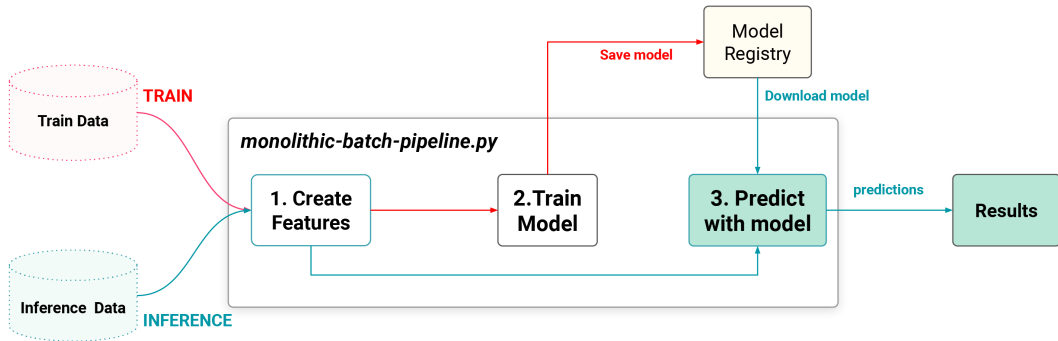
- Notebook and a trained model
=! ML System
- Only when prediction is made
with new data
- For a system we need pipelines



Source: Hopsworks

First Attempt: Monolithic Pipeline

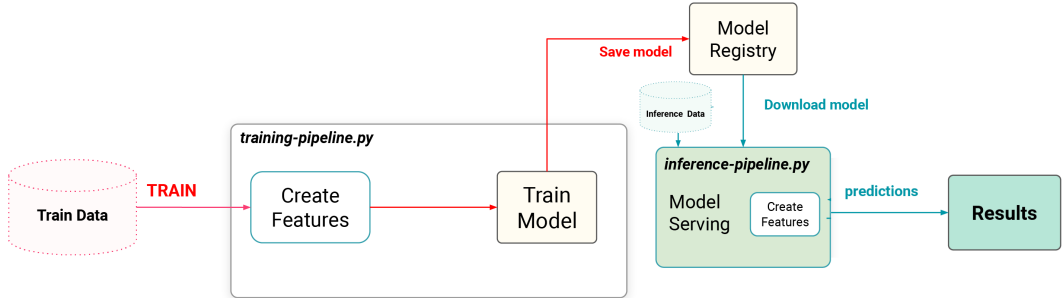
- Everything in one pipeline
- A toggle selects *Training* or *Inference*
- Advantage: Feature calculation occurs only once
- Disadvantage: No modularity, scales poorly, one technology for everything



Source: Hopsworks

Separate Training and Inference

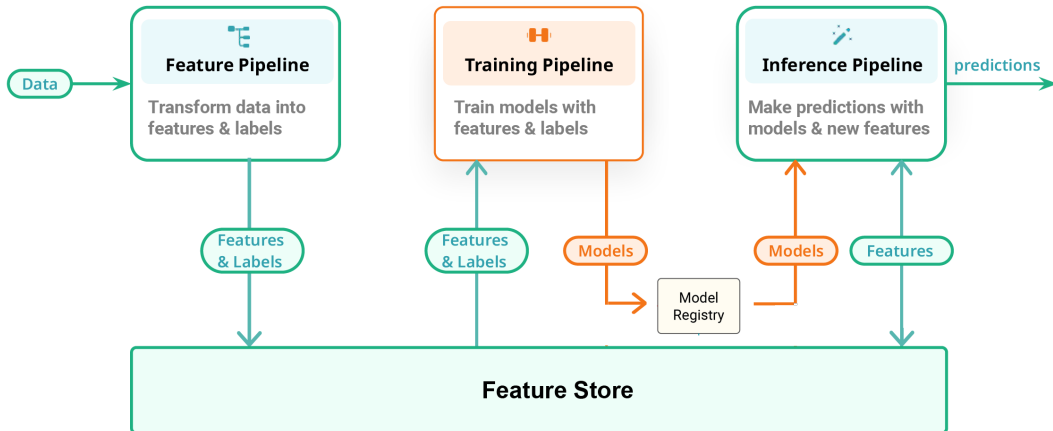
- Natural step: Separate training and inference
- Advantage: More modularization
- Disadvantage: Feature engineering must be duplicated



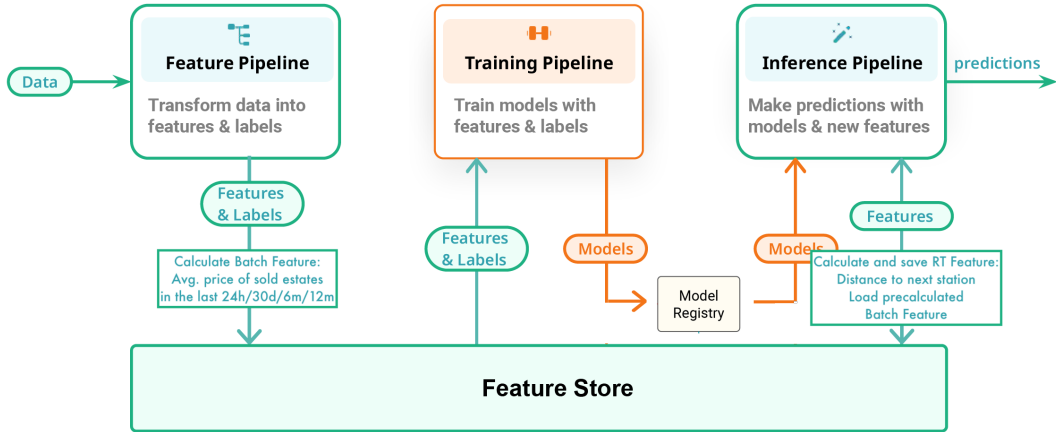
Source: Hopsworks

FTL or 3 Pipeline Architecture

- **Feature Pipeline:** Input = Raw data, output = Features (and Labels)
- **Training Pipeline:** Input = Features and Labels, output = trained model
- **Inference Pipeline:** Input = Current Features+Model, output = prediction



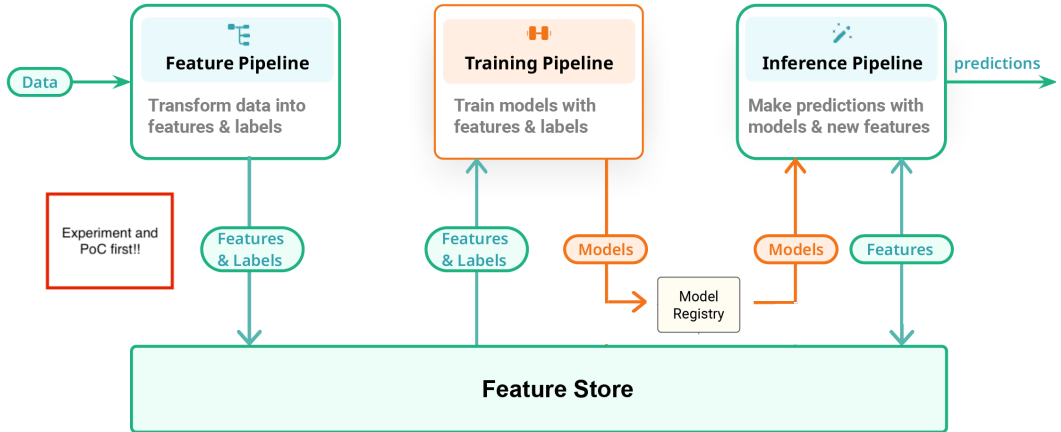
FTI Example



- Technology-agnostic
 - Python/Pandas for training, Spark/dbt for feature calculation
 - Not dependent on a specific ML Registry or Feature Store
 - Cloud or on-premise possible
- Easy to expand
 - Feature Store / Model Registry from simple file storage to full-featured product
- Modular / Reusable / Separately optimizable and scalable

F Pipeline vs. Experimentation Phase

- F before T before I not efficient
- Experiment first, prove that it works or fail fast



- More on monitoring later
- However, a fourth type of pipeline is needed: The Monitoring Pipeline
- Not all monitoring can and should be done in the FTI pipelines
 - Reference data only needed for monitoring
 - Certain amount of data necessary -> does not work within streaming feature pipelines
 - When Natural Labels are strongly delayed -> does not work in I Pipeline